Programming Assignment — Entropy

File name:

Entropy.java

Purpose

This program computes the weighted entropy of one or more attribute distributions. Each input line represents the class distribution for an attribute. The program outputs one line — the weighted entropy for that attribute.

Program Requirements

• Your program must be named exactly Entropy.java.
• It must compile and run as a single Java source file, not part of any package and not dependent on any other source files.
• You may use standard Java library classes and methods (e.g., Math.log, Scanner, etc.), but no external packages.

Compile and run using:
javac Entropy.java
java Entropy < input.txt > output.txt

Input Format

• The first line of the input file contains a single integer N, the number of problem instances.
• Each of the next N lines describes one attribute distribution in this format:

<numClasses>:<counts for value 1>;<counts for value 2>;...;<counts for value M>

Where:
• <numClasses> = number of possible classes (e.g., 2, 3, 4, etc.).
• Each <counts for value i> is a comma-separated list of integers giving the number of instances of each class for that attribute value.
• Different attribute values are separated by semicolons (;).

Output Format

• The program prints N lines, one for each input instance.
• Each output line contains a single floating-point number, representing the weighted entropy of that attribute.
• Round each result to 4 decimal places.

- Sample input and output files will be posted to Moodle.

Notes

• You may assume all input values are valid non-negative integers.
• Use base-2 logarithms.
• The formulas for entropy and weighted entropy are provided in the course slides.
• This program will be autograded by a script. It is important that you adhere to the input and output formats precisely.